

Table D-15. Asthma Prevalence Rates Used to Estimate Asthmatic Populations

Endpoint	Population Group	Asthma Prevalence	Source
Asthma Symptoms	All Ages	7.80%	American Lung Association (2010b, Table 7)
	<5	6.14%	
	<18	9.41%	
	5-17	10.70%	
	18-44	7.19%	
	45-64	7.45%	
	65+	7.16%	
	African-American, <5	9.98%	American Lung Association (2010b, Table 9)
	African-American, 5 to 17	17.76%	
	African-American, <18	15.53%	American Lung Association*
New Onset Asthma**	0-4	3.80%	National Health Interview Survey (2018, Table 4-1)
	5-17	8.93%	
	4-17	8.56%	
	7-21	9.01%	
	Female, 35 to 64	9.90%	
	Female, 65+	10.00%	
Asthma Symptoms (Albuterol Use), Albuterol Use	6-17	9.04%	

* Calculated by ALA for U.S. EPA, based on NHIS data (CDC, 2008).

** Asthma prevalence rates differ for the same age groups due to population weighting

D.8 Other Health Endpoint Occurrence

Baseline incidence estimates for health endpoint occurrences other than a hospitalization or emergency department visit are described below, listed in alphabetical order.

D.8.1 Allergic Rhinitis

Prevalence rates of hay fever/rhinitis are presented by Parker et al. (2009). Parker et al. investigate the associations between long-term ozone exposure and respiratory allergies in children ages 3 to 17 years old. The authors use prevalence data from the NHIS household interview survey and define allergic rhinitis as children with reported hay fever, respiratory allergy, or both within the 12 months prior to the survey. Of the eligible population (72,279), 19.2% of respondents experience allergic rhinitis symptoms within the year prior to the survey, therefore, the national prevalence rate of hay allergic rhinitis is 0.192.

D.8.2 Lung Cancer

The baseline incidence rates for non-fatal lung cancer were calculated using the existent baseline incidence rate for lung cancer *mortality* in combination with the five-year lung cancer survival rate from NCI (2015). We first used the five-year lung cancer survival rate to calculate the total incidence of lung cancer (both fatal and non-fatal) from the baseline mortality rate using the following formula: baseline mortality rate / (1 – five-year survival rate). We then calculated the incidence of non-fatal lung cancer as the difference between total lung cancer incidence and fatal lung cancer incidence (NCI, 2015). Table D-16 presents the baseline incidence of lung cancer mortality, the SEER five-year survival rate, the estimated total lung cancer incidence, and the estimated non-fatal lung cancer incidence rate by age group.

Table D-16. Lung Cancer Incidence Rates

Age Group	Annual Lung Cancer Mortality Incidence [A]	Five-Year Survival Rate [B]	Total Lung Cancer Incidence [C] = [A] / (1 - [B])	Non-fatal Lung Cancer Incidence [D] = [C] - [A]
25-34	0.0000033	34.6%	0.0000050	0.00000175
35-44	0.0000282	34.6%	0.0000431	0.00001492
45-54	0.0002378	22.1%	0.0003053	0.00006746
55-64	0.0007922	20.8%	0.0010003	0.00020805
65-74	0.00019701	21.0%	0.0002494	0.00005237
75-84	0.0032952	14.9%	0.0038722	0.00057695
85+	0.0031820	14.9%	0.0037391	0.00055713

D.8.3 Out of Hospital Cardiac Arrest

The baseline incidence of cardiac arrests occurring outside of the hospital (OHCA) is estimated using the incidence and survival rates reported by Daya et al. (2015). Daya et al. (2015) utilize Resuscitation Outcomes Consortium data to calculate the incidence per 100,000 of OHCA and the survival rate broken down into four age categories, 0 to 17, 18 to 39, 40 to 64, and 65+. We combined the age-specific incidence and survival rates to calculate the baseline incidence for non-fatal OHCA (Table D-17).

Table D-17. Out of Hospital Cardiac Arrest Incidence and Survival Rates

Age	Annual incidence per 100,000 people	Survival Rate	Annual non-fatal incidence per 100 people
0-17	10.1	8.4%	0.008
18-39	33.5	9.8%	0.033
40-64	137.3	14.9%	0.205
65+	553.5	8.8%	0.487

D.8.4 Stroke

We developed non-fatal stroke baseline incidence rates using similar data as was utilized to develop the non-fatal OHCA incidence rates. Yao et al. (2019) provide the annual incidence of stroke in 2013 for individuals over 65 years old, as well as the survival rate, broken down by race, gender, and stroke type (hemorrhagic or ischemic). We combined the incidence and survival rates to calculate the rate of non-fatal stroke by gender and race. We then calculated the overall annual baseline incidence rate of stroke in all individuals over the age of 65 by calculating a weighted average from the stratified. This resulted in a rate of 0.004 strokes per person per year. Table D-18 presents the stratified incidence and survival rates.

Table D-18. Stroke Incidence and Survival Rates

Characteristic	Annual incidence per 100,000 people		Survival Rate		Non-fatal Incidence per 100 people per year			Weight (Study Population)
	Ischemic	Hemorrhagic	Ischemic	Hemorrhagic	Ischemic	Hemorrhagic	Total	
Black Men	551	93	92%	73%	0.00507	0.00068	0.00575	6,155
White Men	407	75	88%	60%	0.00358	0.00045	0.00403	54,079
Black Women	641	94	91%	69%	0.00583	0.00065	0.00648	9,819
White Women	466	77	85%	56%	0.00396	0.00043	0.00439	78,839
Weighted Average							0.00446	

Appendix E. Core Particulate Matter Health Impact Functions in U.S. Setup

In this Appendix, we present the core PM-related health impact functions in BenMAP, i.e., the functions that, as of the current release, U.S. EPA routinely uses in its regulatory analyses. Each sub-section has a table with a brief description of the health impact function and the underlying parameters. Following each table, we present a brief summary of each of the studies and any items that are unique to the study.

Note that Appendix C mathematically derives the standard types of health impact functions encountered in the epidemiological literature, such as, log-linear, logistic and linear, so we simply note here the type of functional form. Appendix D presents a description of the sources for the incidence and prevalence data used in each health impact function.

E.1 Long-term Mortality

There are two types of exposure to PM that may result in premature mortality. Short-term exposure may result in excess mortality on the same day or within a few days of exposure. Long-term exposure over, say, a year or more, may result in annual mortality in excess of what it would be if PM levels were generally lower, although the excess mortality that occurs will not necessarily be associated with any particular episode of elevated air pollution levels. In other words, long-term exposure may capture a facet of the association between PM and mortality that is not captured by short-term exposure. Table E-1 lists the long-term mortality health impact functions.

Table E-1. Core Health Impact Functions for Particulate Matter and Long-Term Mortality

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form
Mortality, All Cause	Di et al.	2017	Nationwide	65-99	O ₃	Annual	0.007046	0.000095	Log-linear
Mortality, All Cause	Turner et al.	2016	Nationwide	30-99	O ₃	Annual	0.005827	0.000963	Log-linear
Mortality, All Cause	Woodruff et al.	2008	Nationwide	0-0		Annual	0.005603	0.004539	Logistic
Mortality, All Cause	Pope et al.	2019	Nationwide	18-99		Annual	0.01133	0.001602	Log-linear
Mortality, All Cause	Wu et al.	2020	Nationwide	65-99		Annual	0.00639	0.000383	Log-linear
Mortality, All Cause	Chen et al.	2023	8 cohorts	25-99		Annual	0.009758	0.002418	Log-linear